

Exploring Semantic Embeddings

Introductory Topics: Text Embeddings, NumPy, Matplotlib, Pickling, and Dimensionality Reduction with PCA and t-SNE

DSC 305

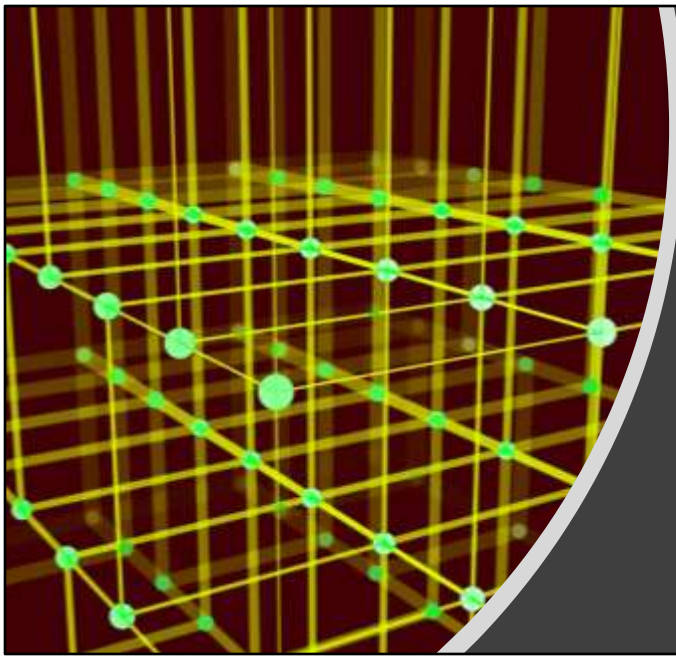
February 20, 2023

The Plan

- Last day of our introductory unit
- Text embeddings
- Feature space (high dimensional)
- Computing distance with NumPy
- Visualizing embeddings with Matplotlib
- Dimensionality reduction with PCA and t-SNE
- Finding neighbors for EDA, classification, and recommender systems

Announcements

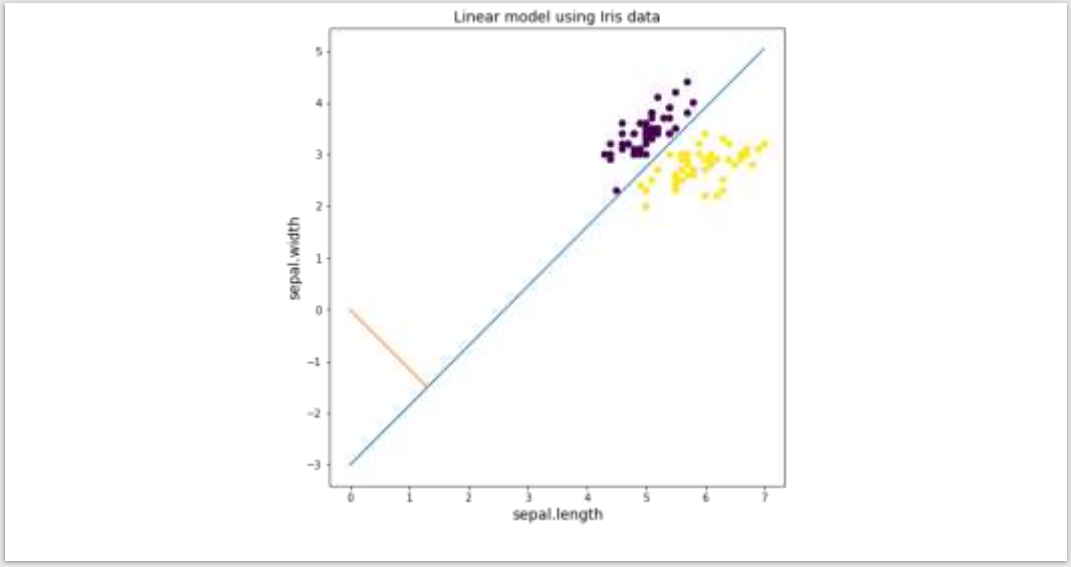
- Quiz 2 on Wednesday
  - Olin 107 Lab
  - Covers everything through today
  - Study guide for QM-2 posted (draft)
  - Can also "level up" on QM-1
  - Work missed questions only
  - Similar, not identical, same numbers
- Continue working on Lab 2 with your partner
- New unit Friday: Tour of Models
  - Logistic regression



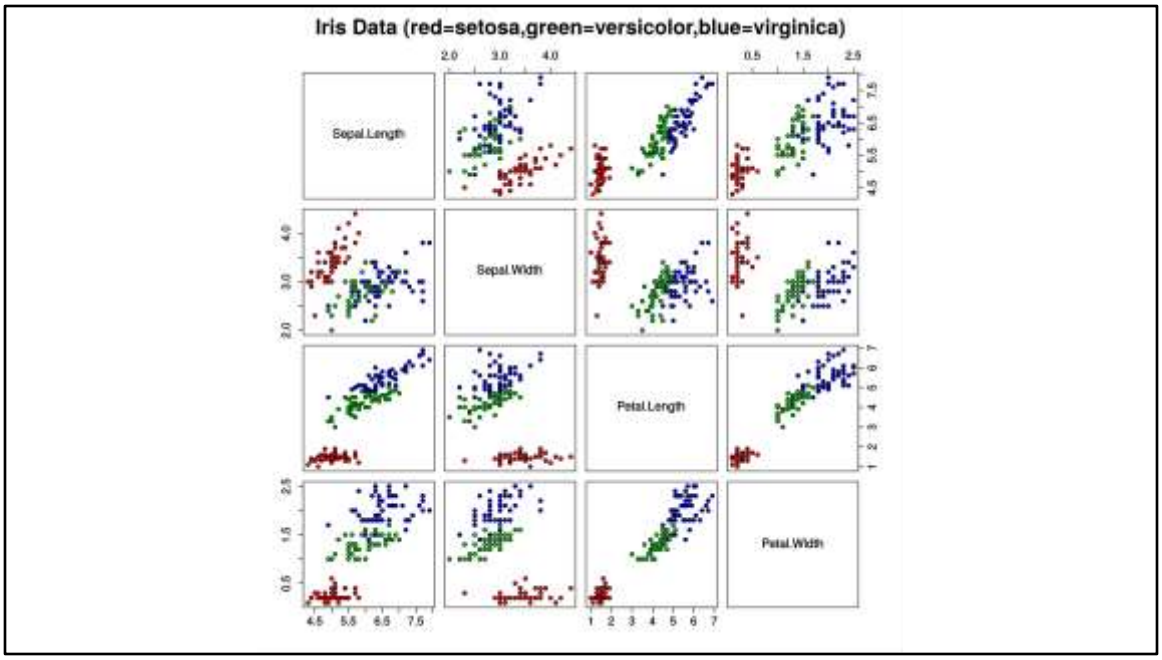
Feature Space

In ML we often embed data in high dimensional space


Iris Data Set



The purple and yellow dots indicate Setosa and Virginica, two varieties of Iris.
Notice how each are clustered on the two sides of this line.
Actually, the data for this dataset is embedded in 4 dimensions because there are 4 features that describe it. And each dot represents an instance.
So this line is actually a hyperplane in 4D space.
It's position is determined by the orange vector that you see here that's orthogonal to it.
In classification learning, we call this hyperplane a decision boundary because just by comparing our data points mathematically
With simple calculations using this orange vector and any data point, we can estimate the probability that the point is a Setosa or a Virginica.


Iris Data (red=setosa,green=versicolor,blue=virginica)
Sepal.Length
Sepal.Width
Petal.Length
Petal.Width

Feature Space

- Important concept in ML.
- Multi-dimensional space.
- Data points as vectors of features.
- Gain insights into the data and make more accurate predictions.
- Linear algebra helpful but does not require deep understanding.
- Can also embed images, sound, and text in feature space.

Feature Space is an important concept in Machine Learning.
It is a multi-dimensional space where data points are represented as vectors and each vector is composed of a set of features.
This allows us to measure the similarity between two data points.
Feature Space can be used to simplify complex datasets and to measure the similarity between data points.
By plotting data points in a Feature Space, we can gain insights into the data and make more accurate predictions.
Feature Space is related to linear algebra but does not require a deep understanding of the subject.


Text Embeddings
Embedding natural language or code in high-dimensional space

Text Embeddings

- Transfer text into a numerical representation.
- Measure similarity between different words.
- Useful for Natural Language Processing (NLP).
- Can be used to classify text and generate recommendations.

Text Embeddings is a technique used to transfer text into a numerical representation. This numerical representation can be used to measure the similarity between different words, phrases, or even whole document. Text Embeddings are useful for Natural Language Processing (NLP) tasks such as text classification, sentiment analysis, and recommendation systems. Text Embeddings can be used to classify text and generate recommendations based on the similarity between words or phrases.

Applications

- **Search** results ranked by relevance to a query
- **Clustering** text grouped by similarity
- **Recommendations** recommend items related to a query
- **Classification** classify text by instances with similar meaning
- **Anomaly detection** identify outliers
- **Diversity measurement** where similarity distributions are analyzed

https://platform.openai.com/docs/guides/embeddings/what-are-embeddings

Text embeddings can be used in a variety of applications, including search, clustering, recommendations, classification, anomaly detection, and diversity measurement. In search, text embeddings can be used to rank results by relevance to a query. In clustering, text embeddings can be used to group text by similarity. In recommendations, text embeddings can be used to recommend items related to a query. In classification, text embeddings can be used to classify text by instances with similar meaning. In anomaly detection, text embeddings can be used to identify outliers. Finally, in diversity measurement, text embeddings can be used to analyze similarity distributions--a measure of the spread of similarity among items. Takeaway: Text embeddings are a powerful tool for NLP tasks, allowing us to measure the similarity between words, phrases, and even whole documents.

Example

- Text: **`The green elephant`**
- Embedding: [-0.0139, -0.015 , 0.014 , -0.0121, -0.0050, ...]
- Space: 1536 dimensions
- Method: autoencoder trained on a large corpus of text (OpenAI text-embedding-ada-002 model)

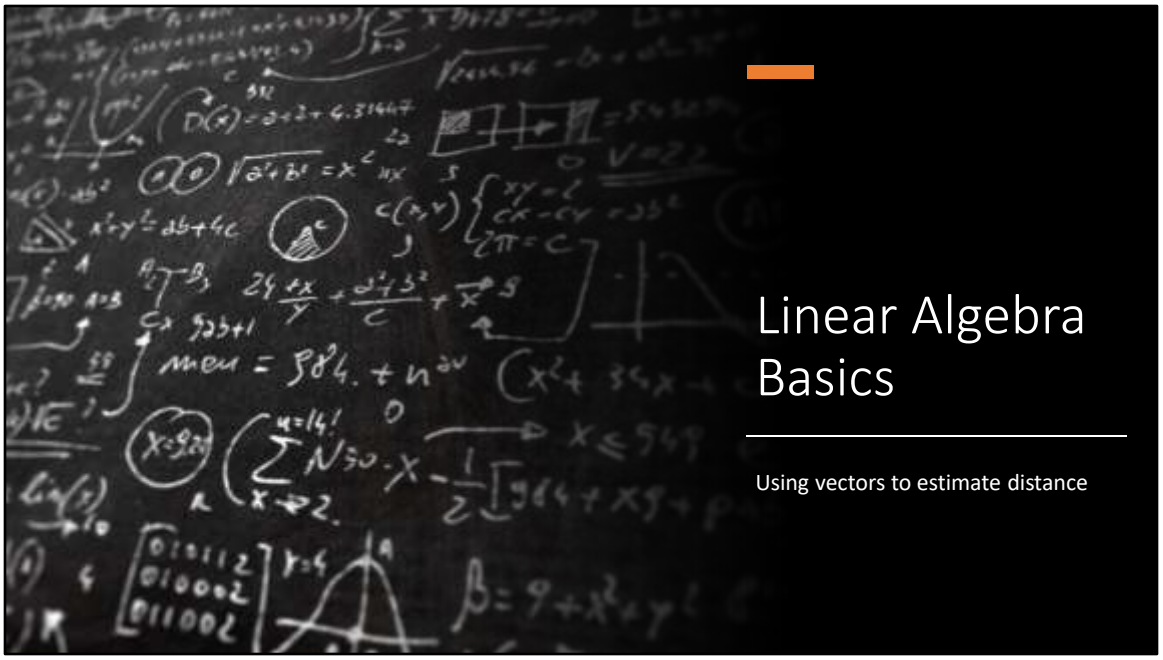
Obtaining embeddings from OpenAI API

Saving embeddings using Python Pickle tool

Coding Demo



Please see the example code posted to Moodle.



Linear Algebra Basics

Using vectors to estimate distance

Dot product

- Calculation used to estimate similarity between 2 vectors.
- Multiplies corresponding entries and adds products together.

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i$$

- u = [3.0, 4.0] v = [11.0, 0.0]. Then $u \cdot v =$ **33**
- a = [2.0, 3.0, 1.0] b = [4.0, 5.0, 2.0] Then $a \cdot b =$ **25**
- Hint: Make sure you can do this for the upcoming test!

Normalization

- **Normalization:** **rescale** vector so its **magnitude** (length) is 1 unit
- **Length of vector:** add square of every element, then take square root
- u = [3, 4] |u| = **sqrt(3*3 + 4*4) = 5**
- v = [11, 0] |v| = **sqrt(11*11 + 0*0) = 11**
- To normalize, divide all elements by length of vector
- u = u / |u| = **(1 / 5) * [3, 4] = [0.6, 0.8]**
- A **unit vector** is a vector with a length of 1.
- All vectors in our text embeddings dataset are unit vectors.

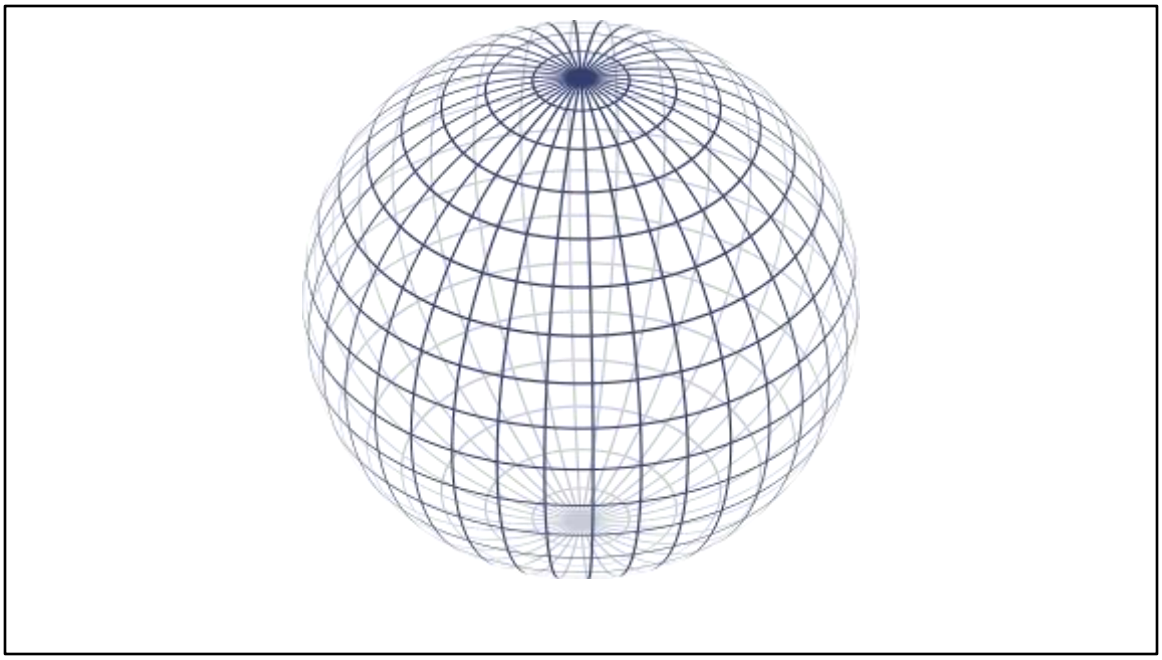
Vector Angle

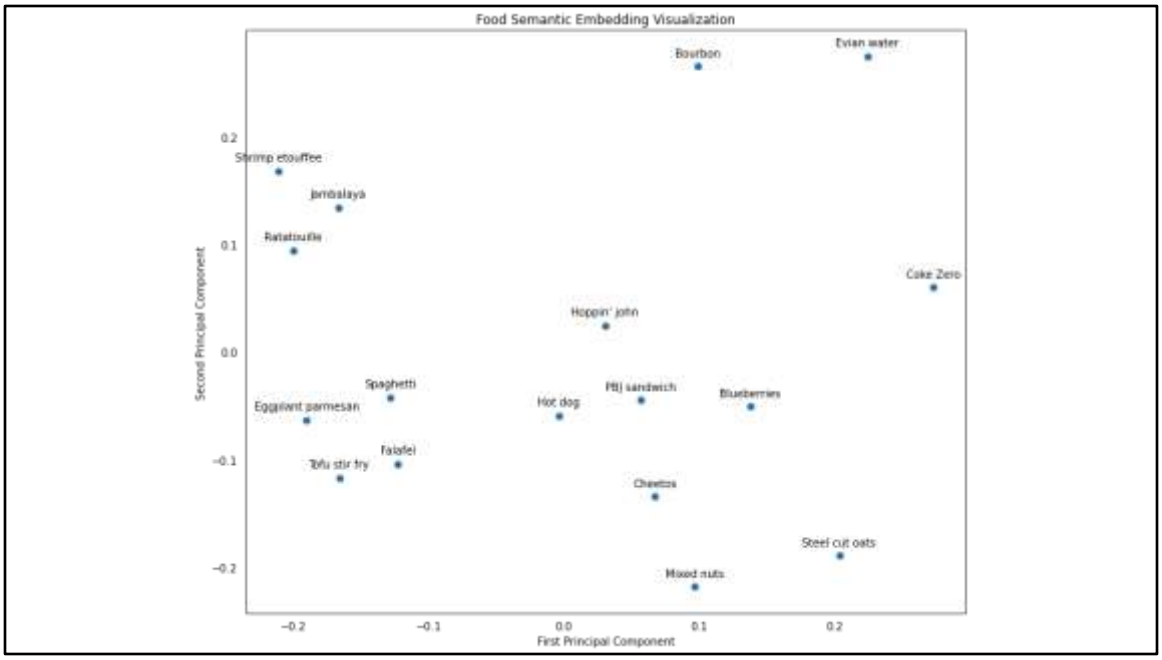
- Formula:

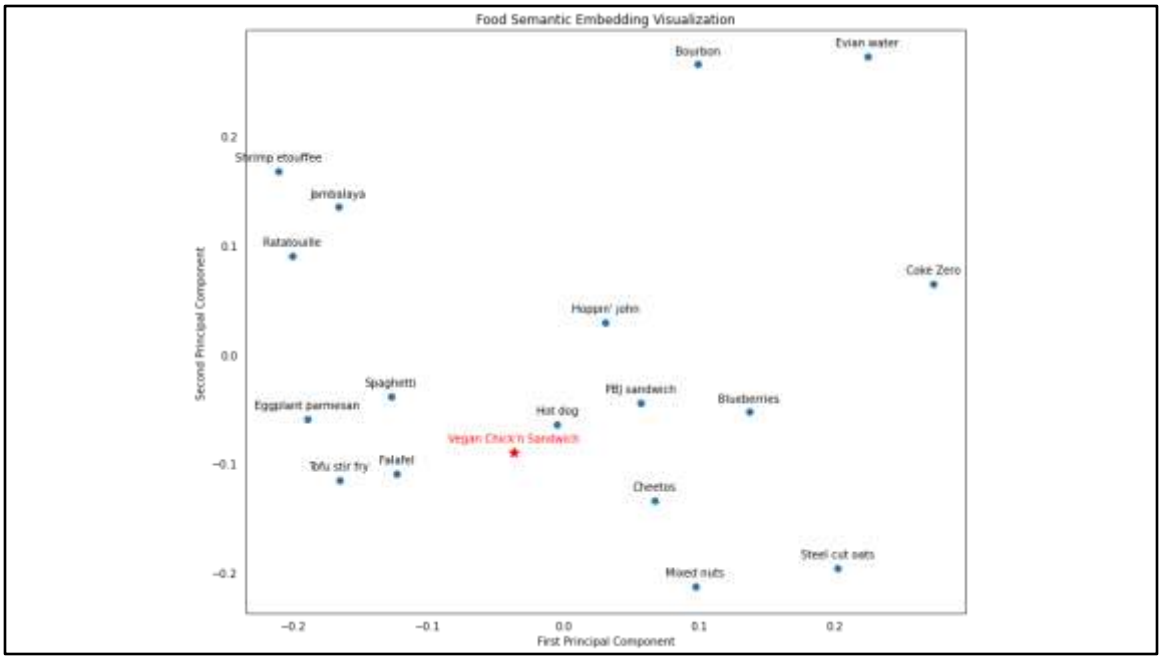
$$\cos\theta = \hat{u} \cdot \hat{v} = \frac{u \cdot v}{\|u\|\|v\|}$$

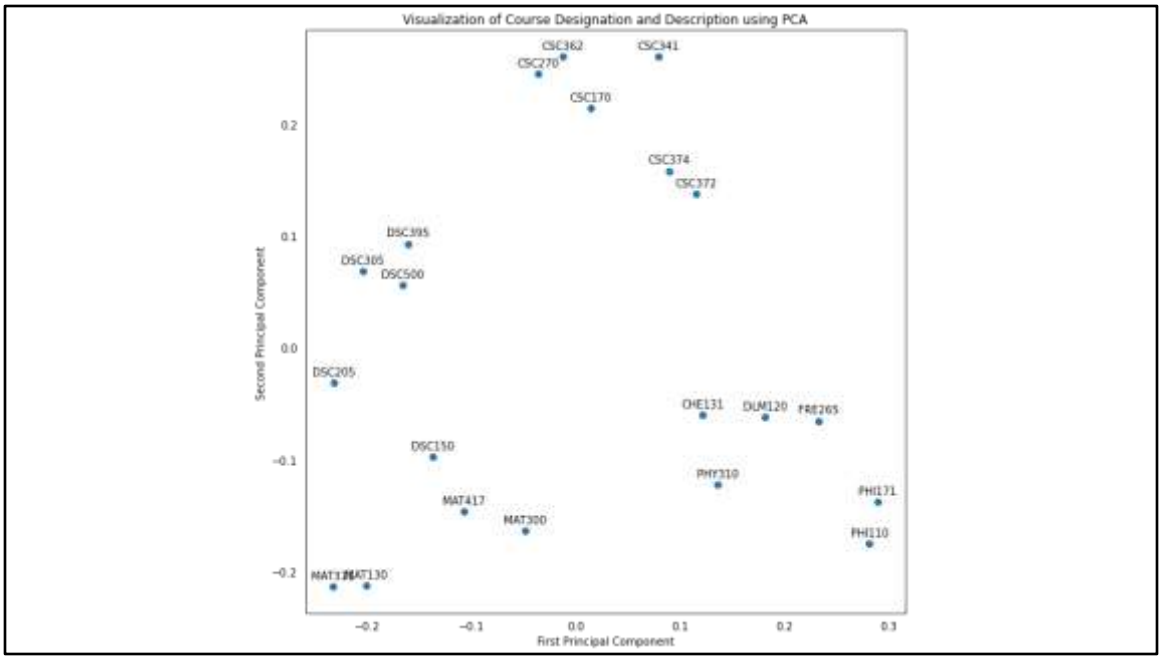
- To find the angle we would have to take the arc cosine.
- But if we just want to **compare** vectors, we just need the **dot product!**
- **The bigger the dot product, the more similar the data points!**
- Dot product of 1 means same point (most similar).
- Dot product of 0 means orthogonal. Can also be negative: [-1, 1]

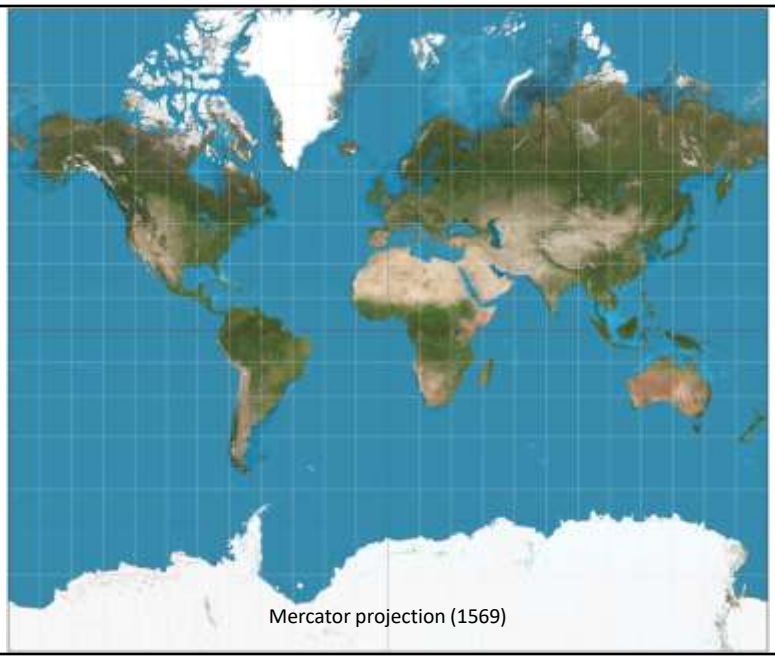

Visualizing
Semantic
Embeddings
Using Projections and Matplotlib




Food Semantic Embedding Visualization
Bourbon
Evian water
Shrimp etouffee
Jambalaya
Ratatouille
Coke Zero
Hoppin' john
Spaghetti
Eggplant parmesan
Hot dog
PBJ sandwich
Blueberries
Falafel
Tofu stir fry
Cheetos
Steel cut oats
Mixed nuts
Second Principal Component
First Principal Component
0.2
0.1
0.0
−0.1
−0.2
−0.2
−0.1
0.0
0.1
0.2


Food Semantic Embedding Visualization
Bourbon
Evian water
Shrimp etouffee
Jambalaya
Ratatouille
Coke Zero
Hoppin' john
Spaghetti
PBJ sandwich
Blueberries
Eggplant parmesan
Hot dog
Vegan Chick'n Sandwich
Tofu stir fry
Falafel
Cheetos
Steel cut oats
Mixed nuts
Second Principal Component
First Principal Component
0.2
0.1
0.0
−0.1
−0.2
−0.2
−0.1
0.0
0.1
0.2

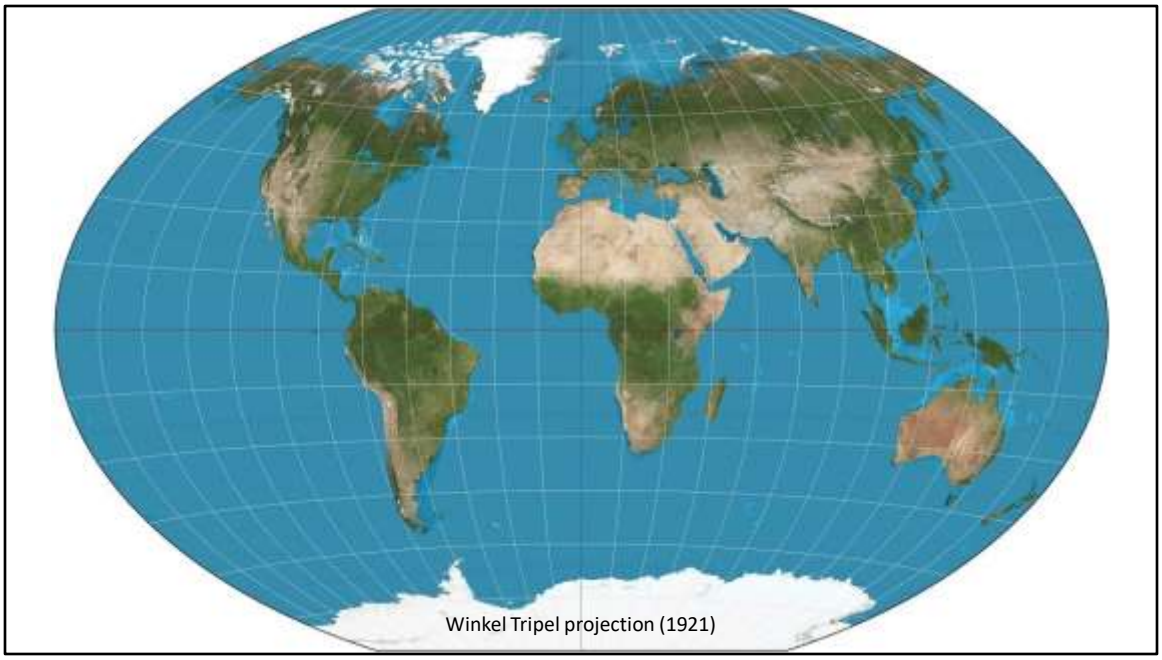

Visualization of Course Designation and Description using PCA
CSC362
CSC341
CSC270
CSC170
CSC374
CSC372
DSC395
DSC305
DSC500
DSC205
CHE131
DUM120
PRE265
DSC150
PHY310
PHI171
MAT417
MAT300
PHI110
MAT130
Second Principal Component
First Principal Component
0.2
0.1
0.0
−0.1
−0.2
−0.2
−0.1
0.0
0.1
0.2
0.3



Mercator projection (1569)

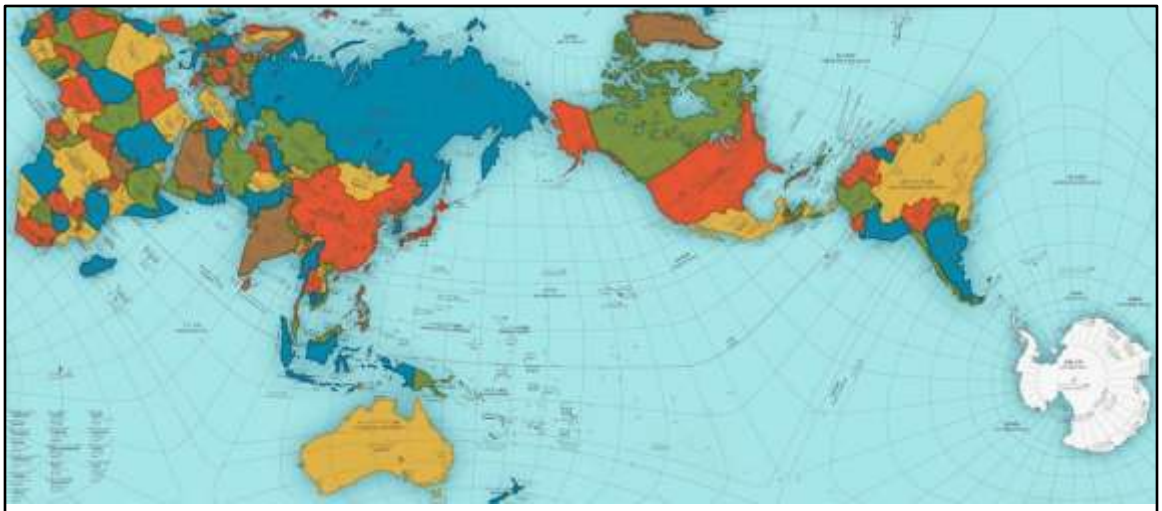
A map projection is a technique used to represent the surface of the Earth on a two-dimensional plane. There are many different types of map projections, each with its own advantages and disadvantages.

The first modern map projection was devised by Flemish cartographer Gerardus Mercator in 1569, known as the **Mercator projection**. This projection is still widely used today, especially for navigation, since it preserves angles and directions. However, the preservation of angles causes the distortion of size, which is why Greenland appears much larger than Africa on the map.



Winkel Tripel projection (1921)

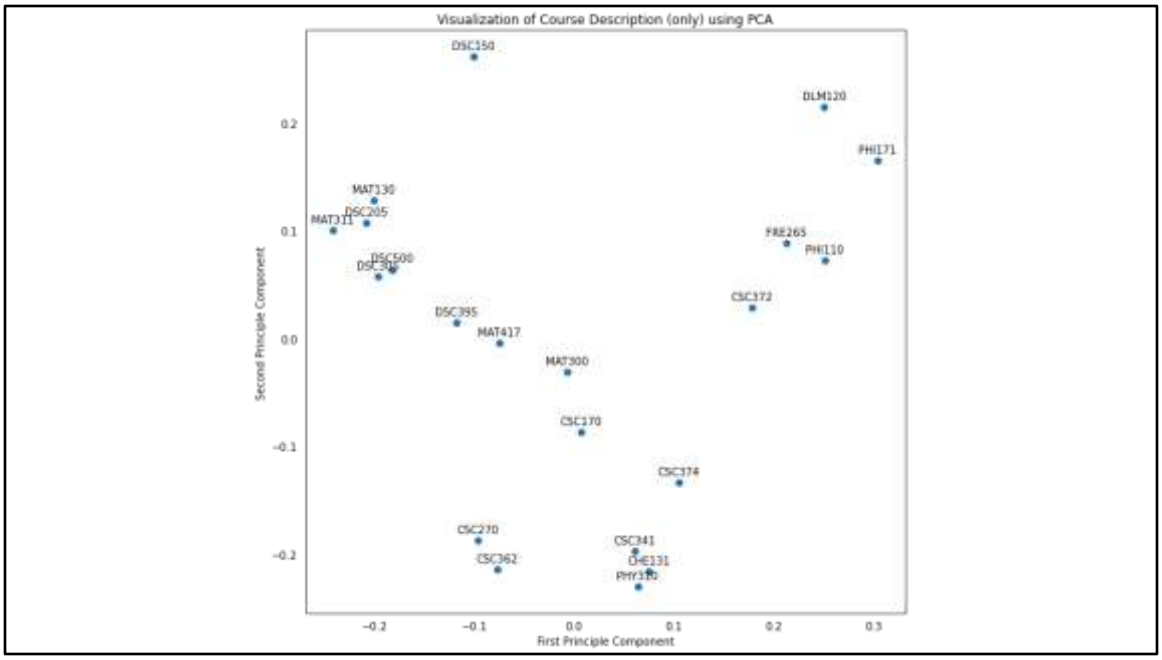
In 1921, German engineer Walter Winkel proposed an improved projection, the **Winkel Tripel**, which reduces the distortion of size. This projection is the most widely used today for world maps.

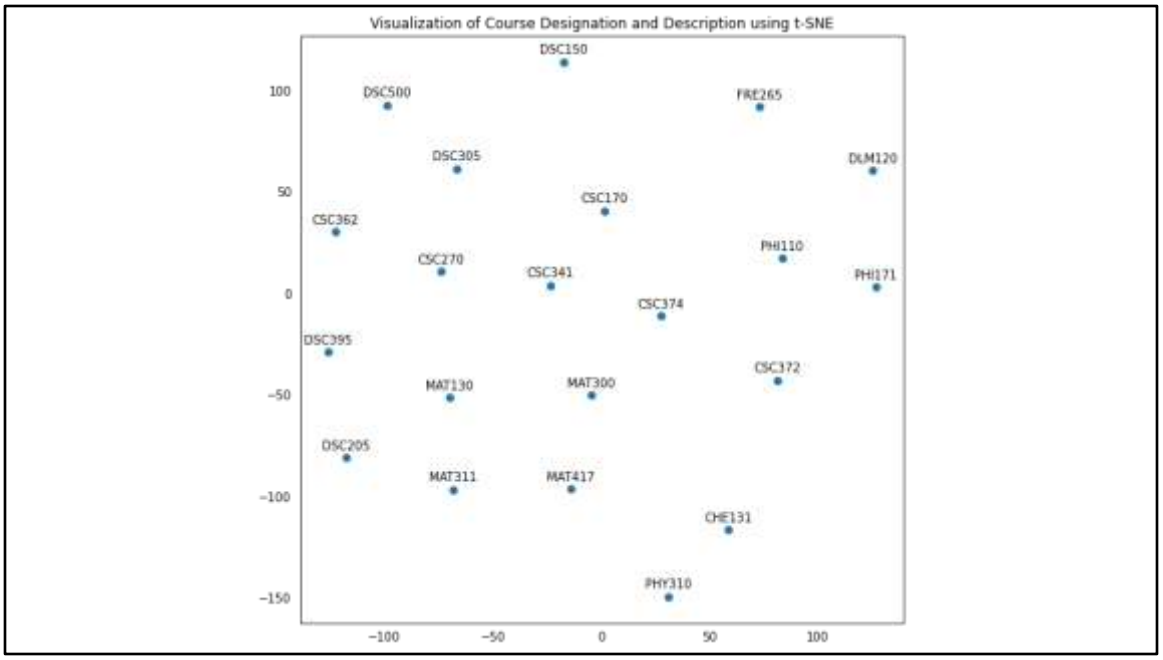


AuthaGraphy projection (Hajime Narukawa, 1999)

https://replogleglobes.com/blog/the-main-types-of-world-map-projections/

The most recent projection, the **AuthaGraphy projection**, was proposed in 1999 by Japanese architect Hajime Narukawa. This projection minimizes distortion of angles, area, and distance. It is used primarily for educational purposes.


Visualization of Course Description (only) using PCA
DSC150
DLM120
PHI171
MAT130
DSC205
MAT311
FRE265
PHI110
DSC500
CSC372
DSC395
MAT417
MAT300
CSC170
CSC374
CSC270
CSC341
CSC362
CHE131
PHY130
Second Principle Component
First Principle Component
0.2
0.1
0.0
−0.1
−0.2
−0.2
−0.1
0.0
0.1
0.2
0.3


Visualization of Course Designation and Description using t-SNE
DSC150
DSC500
FRE265
DSC305
DLM120
CSC170
CSC362
PHI110
CSC270
CSC341
PHI171
CSC374
DSC395
CSC372
MAT130
MAT300
DSC205
MAT311
MAT417
CHE131
PHY310
100
50
0
−50
−100
−150
−100
−50
0
50
100

Projections

- Visualize high dimensional data in 2D or 3D
- Linear: Principal Component Analysis (PCA)
- Non-linear: t-Distributed Stochastic Neighbor Embedding (t-SNE)
- As with map projections, there are advantages and disadvantages
- Sadly, there are always some distortions
  - On previous slide, DSC395 is closest neighbor to CSC372 but looks far away.



Course Recommender System

Example Application

Recommender System

- Predict preferences or ratings that a user would give to an item
- Our example: given **user query**, recommend list of relevant courses
- Simple algorithm:
  - Let u = embedding(user_query)
  - Let v = embedding(course_description)
  - Then $u \cdot v$ denotes similarity of course to user interests
  - Choose the most similar k courses to present to the user
- Many other approaches, e.g., **collaborative filtering** (compare users)

What have we learned?

- Feature space is a central concept in ML
- Text embeddings: geometric distance approximates semantic distance
- If vectors are normalized, bigger dot product suggests more similarity
- Can study geometric relationships with NumPy linear algebraic library
- Can visualize with Matplotlib
- To visualize we first need to project onto 2- or 3-dimensional space
- Recommender system
- Reading check: EMC 1-2, PDSH 1-4 (can skim 2)